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PALEOMAGNETIC DIRECTIONS FROM THE IZTACCIHUATL VOLCANO, MEXICO

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Measurement of partially demagnetized samples from 35 sites above timberline on Iztaccihuatl, a product of late Cenozoic volcanism, revealed entirely normal polarity of the lava flows.

1. Introduction

Iztaccihuatl is located 60 air km southeast of Mexico City (fig. 1). Most of the lavas of this volcano are andesites, though some basaltic flows occur. Iztaccihuatl consists of numerous lava flows from each of several vents, and because the mountain is over 5200 m (17000 ft) high, there is little tree cover to obscure the volcanic stratigraphy. The work reported here is confined to that portion of the volcano above timberline (roughly 4000 m).

The elongate Iztaccihuatl and its conical neighbor to the south, Popocatepetl, make up a mountain range known as the Sierra Nevada, which range forms a part of the eastern border of the Valley of Mexico. The geology and paleomagnetism of the Valley of Mexico are described by Nairn (in preparation). The fancied resemblance of Iztaccihuatl to a reclining woman is responsible for its popular appellation of 'Sleeping Lady' and also for its morphological division into four units – Feet, Knees, Breast and Head – each of which appears to consist of flows from a different vent. The immense height and tropical location of the mountain have assured that it has been subjected to past intense glaciation and more recently to high rainfall. Thus erosional sculpturing renders difficult the interpretation of the volcanic history of the mountain. The most likely interpretation of field observations of relationships between the lava flows is shown in fig. 2. This intepretation is not entirely certain as in some cases lava flows must be projected visually across valleys, a somewhat risky procedure. As shown in fig. 2, the flows decrease in age from south to north with some overlap in age of flows from the Knees and Breast.

2. Sampling

Field work was carried out during the summers of 1967 and 1968. Samples were taken with a gasoline-powered coring device [1] at the more accessible sites and by block sampling techniques at remote sites. Cores were oriented by the method of Helsley [2]. In sampling, an effort was made to take sites at as many flows as possible in chronological sequence. It was not possible to sample some flows because of their inaccessibility, but samples were taken from at least one flow from each vent, and the lack of any evidence of erosion between the flows from any given vent suggests that the flows from each vent were extruded in rapid succession.

As the samples taken in 1967 yielded, when measured, entirely normal polarities, sampling the following summer was devoted largely to flows intermediate in age between those sampled in 1967 in an effort to show that the all-normal result was not an artifact of insufficient sampling. In particular, as the Feet appear far more dissected than the Knees which are stratigraphically immediately above the Feet, and as the results of the first summer showed that the oldest flow of the Feet and the oldest flow of the

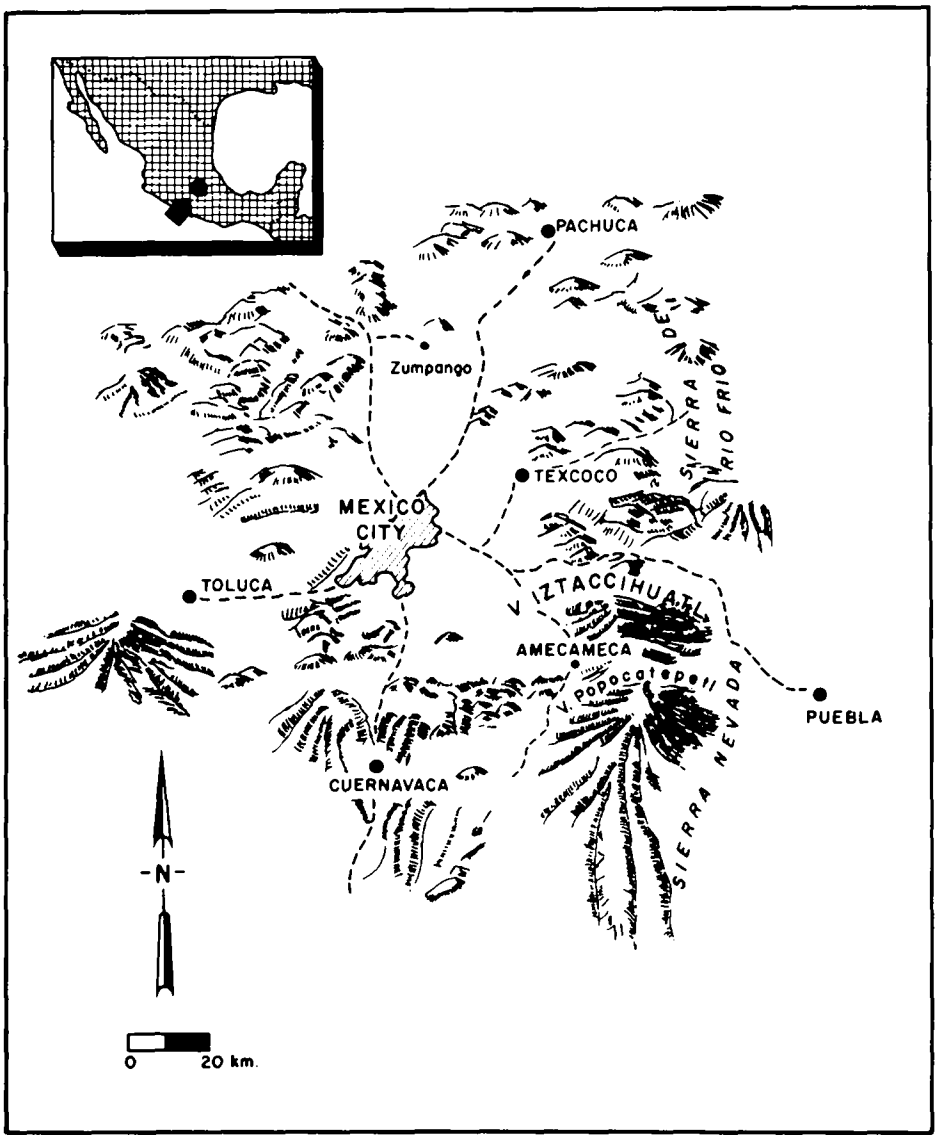


Fig. 1. Location of Iztaccihuatl.

Knees were both magnetized in the normal sense, it appeared that the most likely place where a reversal might have been skipped was between these two flows. Therefore, in the second summer, every flow of the Feet was sampled. In addition, a flow-by-flow collection was made at a small volcano which lies to the south of Iztaccihuatl proper but which rests on the lowest flow of the Feet. Several additional sites were taken both to the south and to the north of the main mountain. The location of all sites is given in fig. 3.

In sampling, seven samples were taken per site, one more than the optimum number of Doell and Cox [3]. As Doell and Cox have shown, statistically the new information obtained from processing each additional sample drops off sharply after six samples.

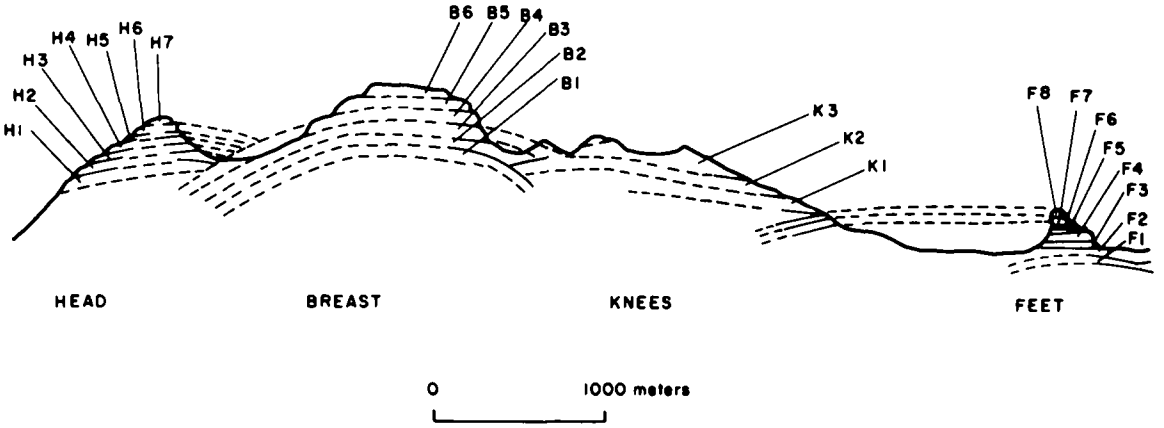


Fig. 2. Idealized cross section of Iztaccihuatl. Numbers refer to individual flows.

3. Laboratory investigation

In the laboratory, cores were taken from the block samples with a trepanning machine [4]. All cores were then cut into as many one-inch specimens as length would permit. For most sites, core lengths were such as to permit one specimen from each of two cores per site to be cleaned by thermal demagnetization and one specimen from each of these same two cores to be cleaned by AF demagnetization.

Thermal demagnetization was carried out in a nonmagnetic furnace in a mumetal box [5], with final cancellation of the geomagnetic field being accomplished with a coil pair surrounding the furnace. The mumetal box effectively reduced diurnal variations in the Earth's field, and specimens were placed sufficiently far apart in the furnace to ensure that each specimen was exposed to a field of no more than 50 γ during demagnetization.

AF demagnetization was also carried out in the mumetal box with an apparatus which rotated the specimen about three axes simultaneously. Alternating fields up to 1200 Oe (peak) were used as needed. The fact that in the majority of cases the directions of remanence obtained after demagnetization by either the thermal or the AF technique were in agreement implies that no spurious components of magnetization were added during either demagnetization procedure.

A typical pair of demagnetization curves is shown in fig. 4. The single Curie point in the vicinity of 550°C found in all the thermal demagnetization curves indicates that an essentially pure magnetite is the principal carrier of magnetism in the Iztaccihuatl rocks. However, reflected light microscopy reveals that hematite and maghemite, both of unknown Ti content, are present in some of the specimens. The steady decrease of magnetization with temperature which begins even at low temperature in fig. 4a suggests multidomain behavior of the magnetic constituent.

Fig. 4b shows that much of the magnetization of these rocks is unstable with respect to AF demagnetization, but that a small and very hard component does exist. Because any changes of direction of the remanence during demagnetization usually occurred only in the steepest part of this curve, the high coercivity component is assumed to reveal the direction of the original TRM of the lava.

The remaining cores, i.e., those not treated by both thermal and AF demagnetization, were subsequently AF demagnetized in those fields judged most effective in removing the secondary component. This judgment was based on the behavior of direction and intensity of remanence during demagnetization of the first two cores from each site.

After demagnetization, measurement of the magnetization of the specimens was carried out on an astatic magnetometer, and later on the fluxgate spinner magnetometer designed by Foster [6] which became available during the progress of the work. Magnetic susceptibility was measured with a Soiltest MS-3A bridge.

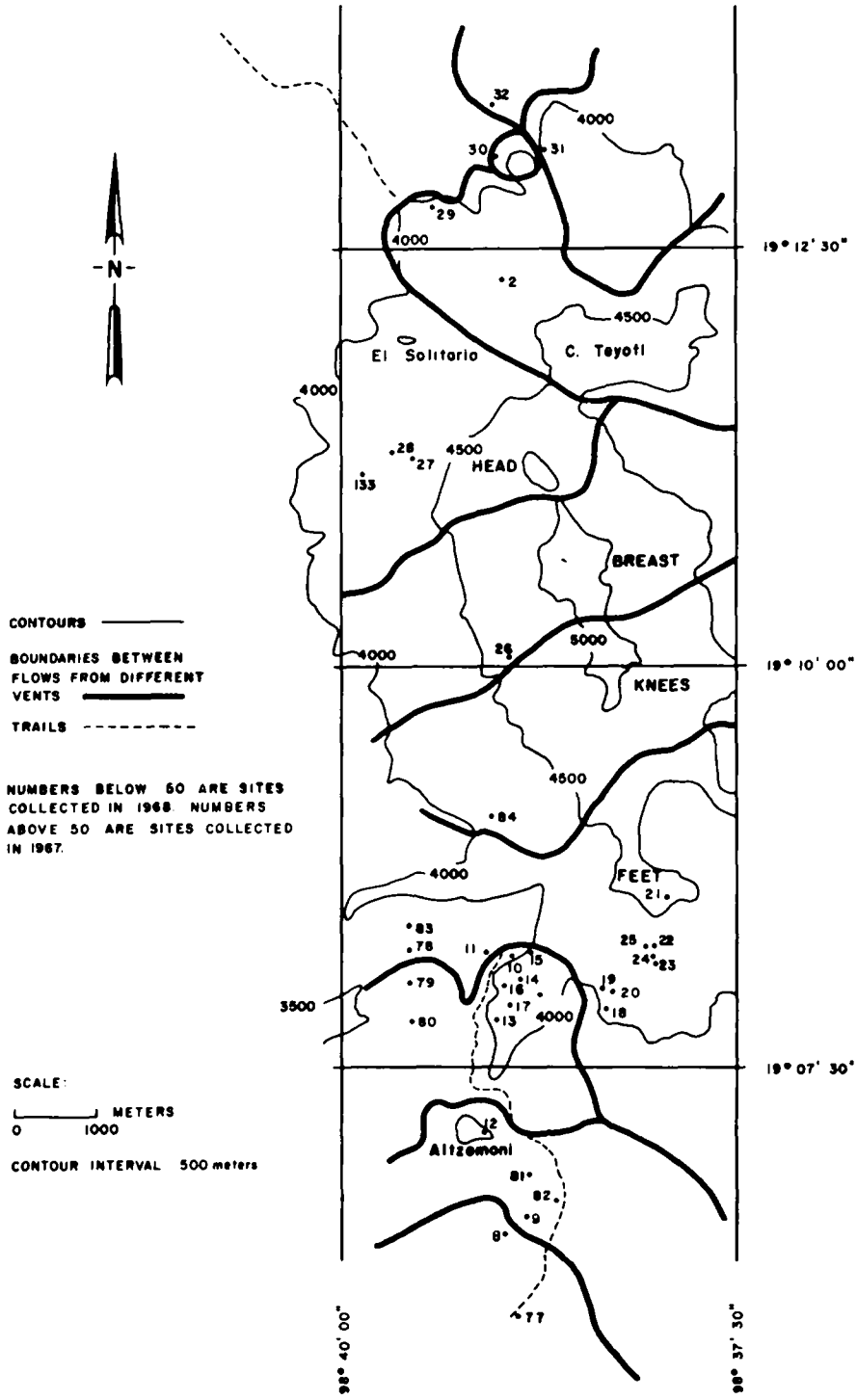


Fig. 3. Location of sites on Iztaccihuatl.

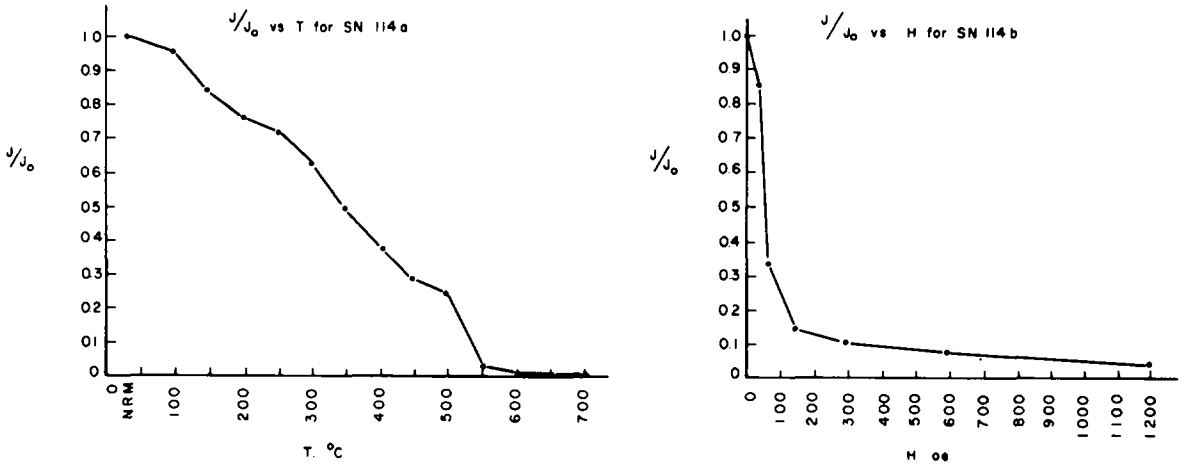


Fig. 4. Demagnetization curves; (a) thermal, (b) AF.

Table 1
K-Ar ages.

Site	Age, my	$^{40}Ar^*/total\ ^{40}Ar$	Reference
Caldera of Iztaccihuatl	5.1 ± 0.5	0.053	Geochron Laboratories #R-0974. Andesite. K-Ar whole rock analysis.
Knees of Iztaccihuatl	13.3 ± 1.2	0.101	Geochron Laboratories #F0973. Andesite. K-Ar. analysis of plagioclase (andesine) concentrate.
Pulpito del Diablo	8.7 ± 0.8	0.190	Geochron Laboratories #A0975. Andesite. K-Ar analysis of hornblende and plagioclase concentrate.

($^{40}Ar^*$ = radiogenic ^{40}Ar)

Constants used: $\lambda_\beta = 4.72 \times 10^{-10}/y$
$\lambda_\epsilon = 0.585 \times 10^{-10}/y$
$^{40}K/K = 1.22 \times 10^{-4} g/g$

$$Age = \frac{1}{\lambda_\epsilon + \lambda_\beta} \ln\left[\frac{\lambda_e + \lambda_\beta}{\lambda_e} \frac{^{40}Ar^*}{^{40}K} + 1\right]$$

For several of the sites, specimens were found to have quite large secondary components. Q_{NRM} (=intensity of NRM/intensity of magnetization induced by the Earth's field) generally was over 100 for these specimens and ranged as high as 1000. The large values of Q_{NRM} and the widely scattered directions of the secondary components for specimens from each of these sites indicate that the secondary components had their origin in intense fields associated with currents from lightning strikes [7].

Table 2
Site parameters.

Site no.	Mean declination of remanence	Mean inclination of remanence	No. of samples	R	k	α_{95}	δ	Demagnetizing field (Oe)	Flow no. or location
1967– 77	354.3	+ 17.6	7	6.20	7.5	23.6	27.6	1200	Paso de Cortes
78	1.3	+ 28.3	7	6.98	243.2	3.9	4.8	300	South of Feet
79	13.8	+ 31.8	4	3.97	96.1	9.4	7.2	600	S1*
80	354.6	+ 33.7	7	6.54	13.0	17.4	21.0	600	SW of Feet
81	346.5	+ 44.8	6	2.93	1.6	81.8	60.7	300	Altzamoni
82	356.1	+ 39.8	8	7.96	178.0	4.2	5.7	600	Altzamoni
83	347.8	+ 32.0	6	5.32	7.3	26.6	27.6	77	Probably F1
84	355.5	– 8.0	7	6.92	70.5	7.2	8.9	77	K1
133	10.2	+ 25.5	6	5.98	207.5	4.7	5.1	600	One of the lower flows of the Head
1968– 2	351.8	+ 21.1	6	2.16	1.3	118.7	68.9	1200	Cerro Teyotl
8	289.6	– 10.9	7	2.80	1.4	88.4	66.4	600	S of Altzamoni
9	10.3	+ 41.0	7	5.93	5.6	28.0	32.2	600	S of Altzamoni
10	7.4	+ 28.5	6	5.97	199.3	4.8	5.2	600	S1
11	3.6	+ 23.5	7	6.96	168.4	4.7	5.8	600	F1
12	12.5	+ 42.7	7	6.08	6.5	25.7	29.8	150	Altzamoni
13	355.9	+ 31.5	7	6.99	569.1	2.5	3.2	1200	S2
14	10.9	+ 45.7	7	6.43	10.6	19.4	23.2	1200	S4
15	127.4	+ 10.0	6	3.39	1.9	68.3	55.6	300	S5
16	13.4	+ 38.6	7	6.81	31.2	11.0	13.5	600	S3
17	21.7	+ 28.2	7	6.32	8.8	21.6	25.5	600	S4
18	17.3	+ 52.9	7	6.91	65.6	7.5	9.3	600	F1
19	20.6	+ 51.0	7	6.81	31.7	10.9	13.4	600	F2
20	22.7	+ 60.8	7	6.96	158.4	4.8	6.0	600	F3
21	352.4	+ 32.1	7	6.95	131.7	5.3	6.5	600	F8
22	354.8	+ 32.9	6	5.22	6.4	28.7	28.6	300	F7
23	0.3	+ 24.6	7	6.86	43.8	9.2	11.4	600	F6
24	355.3	+ 21.3	7	6.93	82.4	6.7	8.3	300	F5
25	27.0	+ 61.5	6	5.98	327.6	3.7	4.1	1200	F4
26	331.9	+ 24.4	7	6.80	29.3	11.3	13.9	300	B1
27	354.0	+ 30.0	6	5.94	79.6	7.6	8.3	600	H7
28	349.5	+ 29.4	5	3.66	3.0	53.7	42.9	1200	H6
29	7.4	+ 38.1	7	6.92	72.6	7.1	8.8	300	Cerro Teyotl
30	356.6	+ 31.9	7	6.92	79.4	6.8	8.4	1200	North of Head
31	3.8	+ 35.9	7	6.91	64.3	7.6	9.4	600	North of Head
32	356.7	+ 30.0	7	6.86	43.2	9.3	11.4	1200	North of Head

* Numbers prefixed by S refer to flows of a small volcano south of the Feet.

4. Age of Iztaccihuatl

On the basis of its highly dissected condition, the Feet volcano is considered by the Mexican volcanologist Mooser [8] to be of Tertiary origin, while the Knees, Breast, and Feet are assigned a birth in the Quaternary. Through Mooser's efforts, three sites from Iztaccihuatl have been dated radiometrically (table 1). In table 1, the caldera and the Pulpito del Diablo sites are located relatively low on the slopes of the mountain, stratigraphically beneath the flows studied in the present investigation, while the site at the Knees corresponds to flow K1 studied here. As the caldera and Pulpito del Diablo sites are both

stratigraphically below the Knees, clearly the K-Ar ages of table 1 are inconsistent. The specimens dated were fresh and in all cases were from locations deep within the flows so that Ar diffusion losses due to reheating by subsequent volcanism would not have occurred. In addition, the rock from the Knees comes from a lofty part of the volcano where erosion is proceeding very rapidly, and thus it seems quite unlikely that this site has persisted for 13 my as indicated by the K-Ar date. Therefore, it appears quite probable that this number for the Knees is in error.

5. Results

The directions of remanence of the specimens were combined by Fisher's [9] methods to yield the following site parameters: the mean declination, the mean inclination, the resultant R of unit vectors representing the sample magnetic directions, the best estimate k of the precision of the distribution, the semivertical angle α_{95} of the cone of 95% confidence, and the angular standard deviation δ. These quantities, along with the number of samples treated from each site, and the peak field in which all samples from the site were treated, are given in table 2.

Table 2 shows that there are no reversely magnetized sites; sites 1967-81, 1968-2, 1968-8, and 1968-15 are ambiguous, and the rest show normal polarity. The ambiguous directions are thought to result from the failure to remove completely the lightning-induced secondary components. Such secondary components are also believed responsible for the large α_{95} for some of the other sites.

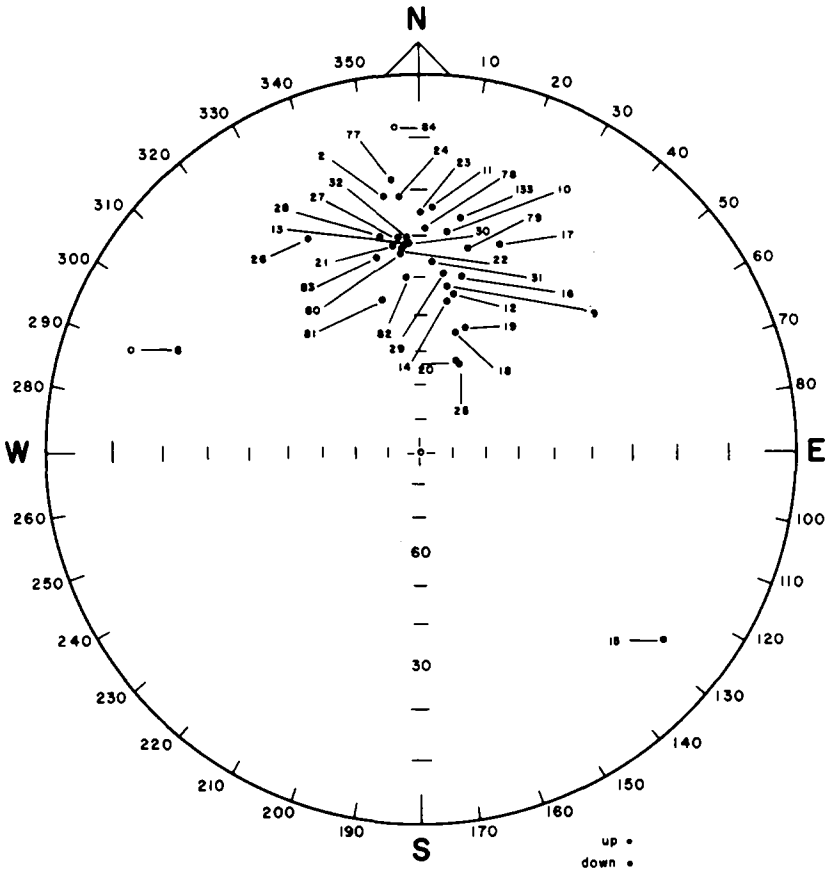


Fig. 5. Site mean directions.

The site mean directions are plotted in stereographic projection in fig. 5.

The mean pole position for the 35 sites is 89°N, 4°E.

6. Conclusions

The portion of the Iztaccihuatl volcano above timberline and other volcanoes immediately to the north and to the south of the main mountain were found to have normal magnetic polarity. These data are consistent with, but not sufficient proof of, Mooser's hypothesis that the Knees, Breast, and Head originated during the Quaternary, a hypothesis which is contradicted by the improbable 13.3 my K-Ar age for the Knees as described above. Whether or not the Knees, Breast, and Head were extruded during the Brunhes normal polarity epoch, the highly eroded state of the Feet volcano suggests that it may have been formed during an earlier epoch of normal polarity than was the rest of the mountain.

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References

[1] R.R. Doell and A. Cox, Palaeomagnetic sampling with a portable coring drill, in: Methods in Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds. (Elsevier, Amsterdam, 1967) 21.

[2] C.E. Helsley, Advantages of field-drilling of samples for palaeomagnetic studies, in: Methods in Palaeomagnetism, D.W. Collinson, K.M. Creer and S.K. Runcorn, eds. (Elsevier, Amsterdam, 1967) 26.

[3] R.R. Doell and A. Cox, The accuracy of the paleomagnetic method as evaluated from historic Hawaiian lava flows, J. Geophys. Res. 68 (1963) 1997.

[4] D.W. Collinson and A.E.M. Nairn, A survey of paleomagnetism, Overseas Geol. Min. Resources 7 (1960) 381.

[5] B.J. Patton and J.L. Fitch, Design of a room-size magnetic shield, J. Geophys. Res. 67 (1962) 1117.

[6] J.H. Forster, A paleomagnetic spinner magnetometer using a fluxgate gradiometer, Earth. Planet. Sci. Letters 1 (1966) 463.

[7] K.W.T. Graham, The re-magnetization of a surface outcrop by lightning currents, Geophys. J. 6 (1961) 85.

[8] F. Mooser, Historia tectonica de la cuencia de Mexico, Boletin de la Asociación Mexicana de Geólogos Petroleros 15 (1963) 239.

[9] R.A. Fisher, Dispersion on a sphere, Proc. Roy. Soc. Lond. A 217 (1953) 295.